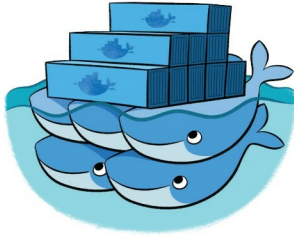


Get Acquainted *with* Docker Swarm

The use of Docker containerisation has become quite common, especially in production. Docker Swarm is a clustering and scheduling tool for Docker containers. With the help of Swarm, sysadmins can deploy and manage a cluster of Docker nodes as a single virtual system.



Docker is a tool that's intended to make the process of creating, deploying and running applications easier by using container based virtualisation technology. Docker is open source container technology that provisions far more apps running on the same old servers compared to traditional VMs.

The Docker engine

The Docker engine is the core component responsible for creating Docker images and running them as services.

The core components of the Docker engine are:

- **The Docker daemon**

This is a continuous running program (daemon) that manages the service and other Docker objects tied to it.

- **The REST API**

This specifies interfaces that programs can use to speak to the daemon and directs what it should do.

- **The Docker client**

This is used to interact with the daemon (Docker command).

Docker networking

Docker networking comes into the picture in large scale, real-time scenarios. Docker networking helps us to share data across various containers.

The host and containers in Docker are linked on the basis of a 1:N relationship, which means one host can command multiple containers.

Cluster management

Before we get into the details of Docker Swarm, let us understand some of the basic concepts of a cluster.

Cluster management is a technique to manage one or more cluster nodes. This is done using a cluster manager and agent. A cluster manager is nothing but a GUI or CLI software.

The clustering tool

A clustering tool is software that manages a set of resources through a single point of command. In our case, these sets of resources are nothing but containers. For example, workload distribution management across a distributed system/cluster is very tedious for large enterprise systems. The clustering tool eases matters by automating this task. The instructor will just specify the details such as the cluster's size, settings and some advanced features. Everything else is taken care of by the clustering tool. Docker Swarm is one such example of a clustering tool for containers.

The need for a container orchestration system

Container management becomes very tough in case of a large scale distributed system which involves hundreds of containers.

A few of the key activities are:

- Scaling the number of containers based on the peak load
- Performing rolling updates for containers
- Performing health checks on containers

Docker Swarm is a cluster management and orchestration tool which is inbuilt in the Docker engine. A Swarm is a cluster of Docker engines, or nodes, with which you deploy services.

You can build this using the SwarmKit (a toolkit to orchestrate the system). Swarm mode can be enabled by either initialising a Swarm or joining an existing Swarm.